

glance, becomes important when you factor in the 1.2 GHz advantage the P4 enjoys. The reason for the Athlon 64 3000+ winning is the excellent bandwidth and low latency offered by the HyperTransport bus used by AMD processors.

This analysis is further confirmed when you compare the older Athlon XP 3200+ and Sempron 3100+. The Sempron 3100+ is a 754-pin processor running at 1.8 GHz and uses HyperTransport, whereas the XP 3200+ is a Socket 'A' processor running at 2.2 GHz, but the memory controller is part of the Northbridge, a setup similar to the one used by Intel. Thanks to this, the Sempron 3100+ leads the Athlon XP 3200+. Note how the 2.53 GHz Celeron D was thrashed by the Sempron 2400+ running at 1.63 GHz, underlining the ageing nature of the Celeron line-up.

ScienceMark 2.0

This benchmark suite consists of numerous tests that involve data calculations such as the mathematically-intense molecular dynamics of the MolDyn test. Through this benchmark, a processor's number-crunching ability comes to the fore. We will now take you through the tests we used within this suite.

Primordia

This test relies mainly on the CPU and, to some extent, on the CPU-memory link. The disk system does not figure in the equation at all. AMD's Athlon 64 3000+ comes out as the best number cruncher in this lot of processors.

Here too, the Athlon 64 3000+ beaters the faster-clocked Pentium 4 3.2 GHz Prescott. The longer



pipelines on the Prescott core are the reason for the performance hit: if your job involves number-crunching varied datasets, such as in the scientific calculations made in this benchmark, processors with shorter pipelines score over processors with high core speeds with long pipelines. (We have not factored the Athlon XP 3200+ here because this processor will no longer be available.)

One interesting thing to note was the difference between the Athlon XP 3200+ and the Sempron 3100+. While in the Content Creation test above, we noted that the Sempron was way ahead of the XP 3200+, the tide reverses drastically when it comes to number crunching. This clarifies two things: if your work involves a lot of high-bandwidth work—using image manipulation applications, 3D rendering, or sound editing—a processor with better CPU-to-memory bandwidth will do well. But if your work pertains to pure number crunching, processors with shorter pipelines and higher clock speeds are better suited to your needs.

The Sempron 3100+ and 2800+,

despite their lower clock speeds, were able to beat the Pentium 4 3 GHz and 2.8 GHz quite easily. Both Intel processors are based on the Prescott core and feature longer pipelines.

MolDyn

Here the results were pretty similar to the Primordia test, confirming AMD's superb data-crunching strength relative to Intel's new Prescott core. The higher-clocked Sempron processors beat the Sempron 3100+ and Athlon 2800+. The Celeron D trailed the Sempron 3000+ by a hefty 55-second margin.

Cipher Test

This test is based on the AES block cipher, formerly known as Rijndael. In this test, the results kept the status quo: the Athlon64 3000+ occupied the first position, while the 3.2 GHz Prescott came in at a respectable second. The rest of the story was very similar, with the Semprons giving their Intel counterparts a hard time.

At the low end of the range, the Sempron 2400+ was still beating the Celeron 2.53 GHz convincingly. In this test, core clock speed and work done per clock matters—hence we observe a scaling of performance.

WinRAR 3.3

The Athlon 64 family made their mark here with the Athlon 64 3000+ taking a convincing lead over Intel's P4 3.2 GHz Prescott. It was interesting to see the Sempron 3100+ beating most of the Pentium 4 series by a considerable margin. This was the first time the Celeron D processor actually beat a Sempron—the Sempron 2500+.

How We Tested Motherboards

The motherboards were segregated first on platform and then on chipsets.

For boards based on Socket 478, we used Intel's Pentium 4 3.2 GHz processor and 1 GB of DDR 400 MHz memory. For the LGA 775 socket, we used a 3.4 GHz EE Processor and 1 GB of DDR 400 or DDR II 533 memory, depending on the motherboard support.

For Socket 939-based AMD motherboards, we used an FX-55 processor; for Socket 754-based boards, we used an Athlon 64 3000+ processor; and for socket 'A', we used an Athlon XP 3200+.

For both these rigs, we used a Gigabyte GeForce 5700 card for AGP-based boards and an MSI 5750-based card for PCIe boards.

We used a SATA 120 GB Maxtor Diamond Max drive on both test rigs. Windows XP with SP1 was our choice of operating system for the test. All other required drivers and patches were loaded before continuing with the tests. The operating system was freshly-

installed for all motherboards and the hard drive was defragmented before testing.

PC Mark 2004: The test mainly comprises applications that are used in day-to-day functioning, and generates a score depending on the performance of the system in each individual test.

3D Mark 03: We used the 2003 version since most of the tests in the new version would be skipped due to lack of hardware support in the GeForce 5700-based cards. However, 3D Mark 2003 scales fairly well and was therefore used for gauging the graphics subsystem of the motherboards.

CineBench 2003: This tool is set to deliver accurate benchmarks by testing the computer's raw processing speed, and other areas that affect system performance such as OpenGL, multithreading, multiprocessing, and Intel's new HT Technology.